

Multimodal Foundations

DATA 443/643 · Week 4, note 4.1

After CMU 11-777: five jobs, then joint versus coordinated.

Lecture plan

- 1 Why more than one stream
- 2 Five jobs: represent, align, fuse, translate, co-learn
- 3 Joint concat versus two towers and a cosine
- 4 A 2-D retrieval numeric

Key goal: leave knowing CLIP scores pairs; it does not fuse pixels with token ids.

First time: what / why / formula

- What / why. Five jobs (Baltrušaitis et al.): represent, align, fuse, translate, co-learn. A model can score a pair (CLIP) without decoding a sentence. Pixels, text, and audio are different sensors. Concatenation is one fusion; it is not the only job.
- Architecture. Separate encoders per modality, then a fuse (concat, add, or cross-attention) or a score (cosine).
- Formula. Coordinated: $s(i, t) = \cos(f(i), g(t))$. Joint concat: $[f(i); g(t)] W$.
- Tradeoffs. + You can reuse a frozen encoder. – Concat grows width; alignment needs paired data; “multimodal” without naming the job is not a project.

How. Pick the job first. Retrieval needs a score. Captioning needs a decoder. Classification can be a linear head on a frozen encoder.

Five jobs, not one architecture

- Represent: one vector, or two towers?
- Align, fuse, translate, co-learn are different verbs.
- Captioning maps. CLIP scores.

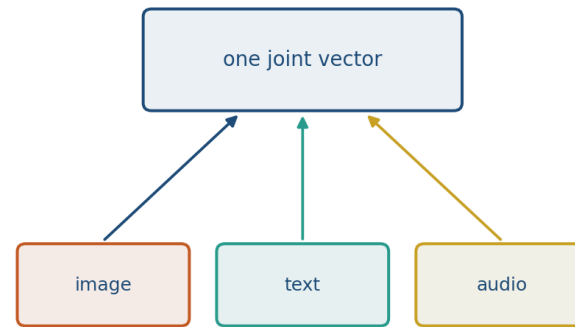
Five jobs. CLIP scores; captioning translates; concat fuses.



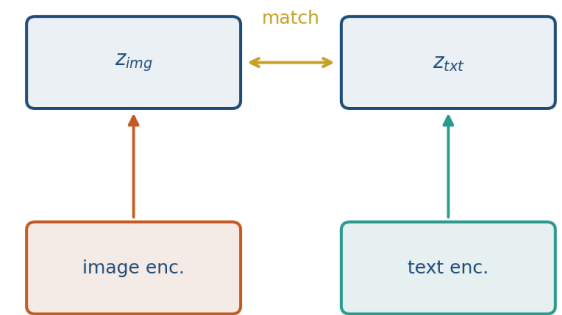
Joint versus coordinated

- Joint: mash streams into one vector.
- Coordinated: a tower each, then a similarity.
- Missing a stream hurts the joint model.

Joint: fuse into one space



Coordinated: keep towers, pull them together



What your demo is allowed to do

Typed search over photos

Coordinated. Embed the query alone. Rank by cosine.

Must output a sentence

Translation. You need a decoder (BLIP, an LLM). Cosine is not a caption.

Rank by cosine

- Match $v = [1, 0]$, $t = [0.8, 0.2]$ is about 0.97.
- Mismatch with $[0, 1]$ is 0.
- Concat of those two vectors is 4-D and needs both.

Coordinated retrieval ranks by cosine. Joint concat needs both streams.

pair	vectors	cosine
match	$v = [1, 0]$, $t = [0.8, 0.2]$	0.97
mismatch	$v = [1, 0]$, $t = [0, 1]$	0

What to take from this lecture

Do not concat and call it CLIP

CLIP never concatenates pixels with token ids in pretraining.

Name the job

Search, caption, or detect a region?
Different boxes on the five-job strip.

Week 4.3

The coordination loss is InfoNCE.
Zero-shot is a text prompt.